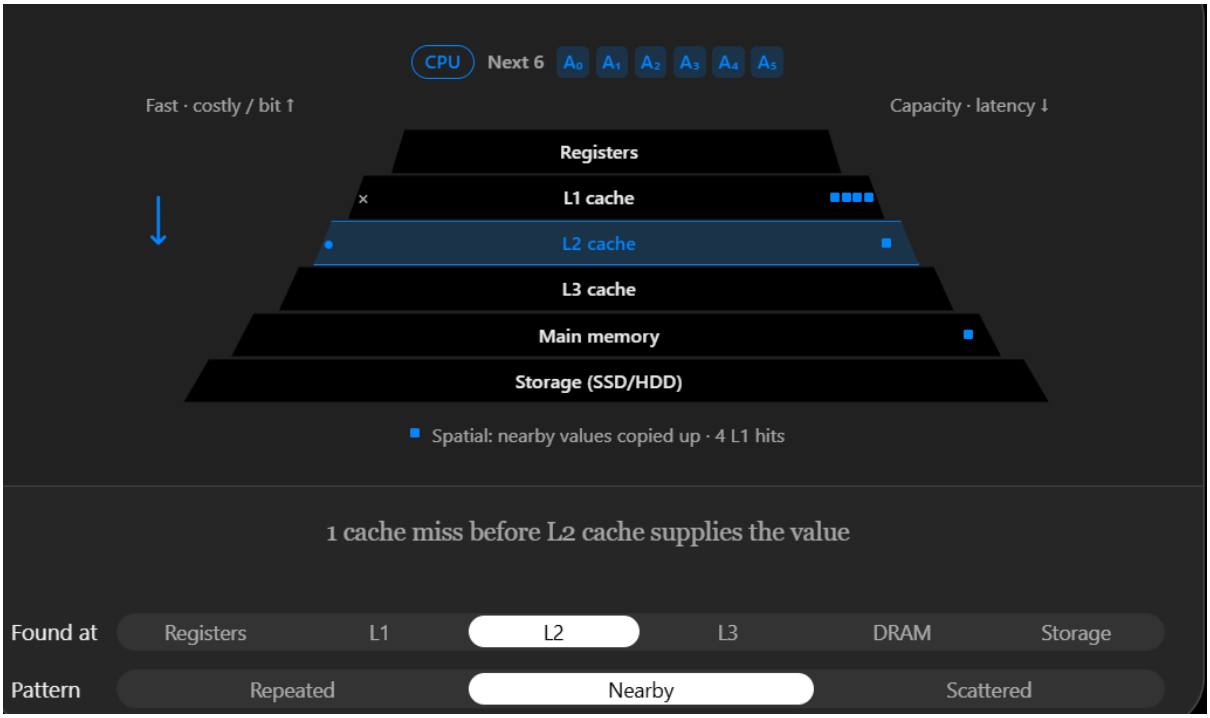
CSA04-Operating Systems

TASK 9 – Evaluate the Memory Hierarchy and I/O Systems

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1. Levels of Memory Hierarchy

Memory hierarchy organizes storage devices based on speed, capacity and cost. Faster memory is generally smaller and more expensive.



Memory Level	Typical Access Time	Typical Capacity
Registers	Less than 1 ns	Bytes to KB
Cache	1–20 ns	KB to tens of MB
Main Memory (RAM)	50–100 ns	GB
SSD	50–200 µs	Hundreds of GB to TB
Hard Disk (HDD)	5–15 ms	TB
Network Storage	1–100 ms or more	TB to PB

Note: These are approximate values and depend on hardware and network conditions.

2. Storage of Hospital Data

Different types of hospital data should be stored at suitable memory levels.

Data Type	Suitable Storage	Reason
MRI Images	SSD and Network Storage	Large files need fast access and long-term storage.
Patient Records	SSD and RAM Cache	Quick retrieval and efficient access.
Monitor Readings	RAM and SSD	Fast processing and permanent recording.
Backup Data	HDD / Network Storage	Large storage capacity and cost efficiency.

Evaluation:

- MRI images require high-capacity storage and fast transfer.
- Patient records need quick access and reliable storage.
- Monitor readings require immediate processing and continuous recording.
- Backups should be stored separately to prevent data loss.

3. Programmed I/O, Interrupt-Driven I/O and DMA

Programmed I/O:

- CPU continuously checks the device status.
- CPU transfers data between the device and memory.
- It wastes CPU time during waiting.

Interrupt-Driven I/O:

- The device sends an interrupt when it needs CPU attention.
- CPU handles the request and transfers data.
- It reduces continuous CPU checking.

Direct Memory Access (DMA):

- DMA transfers data directly between the device and main memory.
- CPU only initializes the transfer and handles completion notification.
- It reduces CPU workload and supports high-speed data transfer.

Comparison

Method	CPU Usage	Speed	Suitable Application
Programmed I/O	High	Low	Simple devices
Interrupt-driven I/O	Medium	Moderate	Keyboard and sensors
DMA	Low	High	MRI scanner

Evaluation: DMA is most suitable for the MRI scanner because it transfers large image files directly into memory without requiring the CPU to handle every data unit.

4. Effect of Caching on Record Look-up

Caching improves the speed of patient record retrieval.

- Frequently accessed patient records are stored temporarily in RAM cache.
- When the same record is requested again, it can be retrieved from cache instead of SSD.
- Cache hits reduce access time.
- Cache misses require fetching data from slower storage.
- It improves database performance and reduces disk I/O operations.

Example: If a doctor repeatedly accesses a patient's medical history, caching allows faster retrieval and reduces database response time.

5. Alternative Concept and Complexity Analysis – Interrupt-Driven I/O vs DMA

Consider an MRI scanner transferring a 500 MB image.

If interrupt-driven I/O generates one interrupt for every 4-byte word:

Step 1: Calculate the number of words

Assume 1 MB = 1,048,576 bytes.

$$500 \times 1024 \times 1024 = 524{,}288{,}000 \text{ bytes}$$

$$\text{Number of Interrupts} = \frac{524{,}288{,}000}{4}$$

$$= \boxed{131{,}072{,}000 \text{ interrupts}}$$

Step 2: Calculate CPU time

Assume each interrupt requires 1 microsecond of CPU processing time.

$$131{,}072{,}000 \times 1 \text{mu s}$$

$$= \boxed{131.072 \text{ seconds}}$$

Thus, the CPU spends approximately 131 seconds handling interrupts alone.

Comparison of Both Methods

Factor	Interrupt-driven I/O	DMA
Interrupts	131,072,000	Usually one or a few per transfer
CPU workload	Very high	Very low
Data transfer	CPU handles each word	DMA transfers blocks
Memory buffer	Small temporary word/buffer	Large block buffers
Performance	Slower	Faster

Memory Buffers Required

Interrupt-driven I/O:

- Uses small temporary buffers for individual words or batches.
- CPU repeatedly handles data and interrupts.
- Large transfers create significant CPU overhead.

DMA:

- Uses larger memory buffers to store data blocks.
- DMA transfers data directly into these buffers.
- Double buffering can be used so one buffer receives data while another is processed.

Evaluation: DMA is clearly better for transferring 500 MB MRI images because it reduces interrupt overhead, CPU workload and transfer delays.

6. Conclusion

1. Memory hierarchy provides different levels of storage based on speed and capacity.
2. MRI images should use SSD and network storage, while patient records need fast database access.
3. Monitor readings should be processed in RAM and stored permanently.
4. Caching improves patient record retrieval speed.
5. DMA is more efficient than interrupt-driven I/O for large MRI image transfers.

Result: The memory hierarchy and appropriate I/O techniques improve hospital system performance, data accessibility and efficient CPU utilization.